

ASSIGNMENT: IP ROUTING AND ROUTING PROTOCOLS

Course:	Computer Networks & IP Routing
Topic:	IP Routing, Subnets, Routing Tables & OSPF

Task 1: Simple Network Diagram & IP Addressing

This network topology illustrates a home local area network (LAN) where multiple consumer devices connect through a central wireless router. The router acts as the default gateway connecting the local private subnet to the external network (Internet).

Network Topology Description

The network consists of three end-user devices connected to a single Wi-Fi router (Home Router). Each device belongs to the same local subnet (192.168.1.0/24), allowing direct Layer 2 communication within the local area network, while traffic destined for external networks is forwarded via the router's LAN interface.

Device Name	Device Type	Interface	IP Address
Smartphone	End Device	Wi-Fi (wlan0)	192.168.1.10 / 24
Friend's Laptop	End Device	Wi-Fi (wlan0)	192.168.1.15 / 24
Smart TV	End Device	Ethernet (eth0)	192.168.1.50 / 24
Router (Gateway)	Routing Device	LAN Interface (br0)	192.168.1.1 / 24

Task 2: Manual IP Routing Simulation Across Subnets

To understand how packets traverse different subnets, we simulate sending a packet from a laptop on Subnet A to a friend's computer located on a separate Subnet B.

Scenario Parameters

Source Laptop IP: 192.168.10.50 (Subnet Mask: 255.255.255.0)

Destination Computer IP: 192.168.20.75 (Subnet Mask: 255.255.255.0)

Gateways / Routers: Router A (Gateway for Subnet A): 192.168.10.1 | Router B (Gateway for Subnet B): 192.168.20.1

Step-by-Step Packet Journey & Routing Table Checks

Step 1: Source Subnet Determination & Gateway Lookup — The source laptop (192.168.10.50) prepares the data packet destined for 192.168.20.75. The laptop performs a bitwise AND operation between its own subnet mask (255.255.255.0) and the destination IP (192.168.20.75) and compares

it with its local network address (192.168.10.0). Since the destination is on a different subnet, the laptop realizes it cannot deliver the packet directly via Layer 2 ARP broadcast. Instead, it must forward the packet to its default gateway (Router A at 192.168.10.1).

Network Destination	Netmask	Gateway	Interface	Metric
192.168.10.0	255.255.255.0	On-link	eth0	1
0.0.0.0 (Default)	0.0.0.0	192.168.10.1	eth0	10

Step 2: Arrival at Router A (First-Hop Router) — The packet arrives at Router A's ingress interface (eth0 connected to 192.168.10.1). Router A strips the Ethernet frame, inspects the Layer 3 IP header, and checks its routing table for the best match for 192.168.20.75.

Network Destination	Netmask	Next Hop / Gateway	Interface	Metric
192.168.10.0	255.255.255.0	Directly Connected	eth0	0
192.168.20.0	255.255.255.0	10.0.0.2 (Router B)	eth1	1

Step 3: Intermediate Hop & Forwarding — Router A matches 192.168.20.75 against the subnet route 192.168.20.0/24. It decrements the Time to Live (TTL) in the IP header by 1, encapsulates the packet into a new Layer 2 frame, and forwards it across the intermediate link to Router B (next hop IP: 10.0.0.2).

Step 4: Arrival at Router B & Final Delivery — Router B receives the packet on its WAN/uplink interface, inspects the destination IP (192.168.20.75), and checks its routing table. It finds a directly connected route for 192.168.20.0/24 on its local interface (eth0). Router B broadcasts an ARP request (if MAC is not cached) to resolve the friend's computer MAC address, encapsulates the packet in an Ethernet frame, and delivers it directly to the destination computer.

Task 3: Zomato-Style Delivery Network Routing Table

Consider a fictional multi-city food delivery network (Zomato-style architecture) operating across three major urban branches in different cities. Each city operates on a distinct subnet connected through a central core/hub router.

Branch / City	Network Address	Subnet Mask	Next Hop IP	Interface
Mumbai Central Hub	10.100.0.0	255.255.0.0	Directly Connected	eth0 (Core)
Delhi Branch	10.101.0.0	255.255.255.0	10.100.0.2	eth1 (WAN-North)
Bangalore Branch	10.102.0.0	255.255.255.0	10.100.0.3	eth2 (WAN-South)

Task 4: Static Routing vs. Dynamic Routing Protocols

Routing methodologies are broadly divided into static routing (manually configured paths) and dynamic routing (protocols that automatically discover network topology and compute optimal paths). Below is a comparative analysis with real-world applications in food delivery and logistics services.

Static Routing

Definition: Static routes are manually entered into a router's routing table by a network administrator. They remain unchanged unless manually modified.

Pros:

1. Predictability & Security: Traffic follows a strict, predetermined path without vulnerability to routing protocol spoofing.
2. Zero Overhead: Consumes zero CPU, memory, and bandwidth resources for route calculation or periodic hello packets.
3. Simplicity in Stub Networks: Highly reliable and easy to maintain for single-connection branch offices.

Cons:

1. Lack of Scalability: Impossible to manually update routing tables across large-scale networks with hundreds of nodes.
2. No Fault Tolerance: If a static link fails, traffic drops entirely; routers cannot automatically reroute around failures.
3. High Administrative Burden: Requires constant manual intervention whenever network topology or IP subnets change.

Dynamic Routing Protocols (e.g., OSPF, RIP)

Definition: Routers running dynamic routing protocols communicate with neighboring routers, exchange topology state information, and automatically calculate optimal paths using algorithms like Dijkstra's SPF.

Pros:

1. Automatic Adaptation & Fault Tolerance: If a delivery hub's primary fiber link is cut, the protocol instantly recalculates and reroutes traffic via alternate paths.
2. Scalability: Seamlessly accommodates thousands of subnets and routers without manual administrator intervention.
3. Load Balancing & Optimization: Automatically chooses the fastest path based on real-time metrics (bandwidth, latency).

Cons:

1. Resource Consumption: Requires router CPU and memory overhead to maintain topology databases and run route calculation algorithms.
2. Configuration Complexity: Setup and troubleshooting require advanced networking expertise and careful metric tuning.
3. Security Risks: Vulnerable to route poisoning, unauthorized router neighbor injections, and convergence delays during link flaps.

Real-World Application Example: Food Delivery App (Zomato / Swiggy)

Consider a high-availability food delivery platform operating hundreds of dark kitchens, micro-fulfillment centers, and regional server nodes. If the platform relied solely on static routing between

regional fulfillment servers and payment gateways, a single fiber cut between Mumbai and Bangalore would completely sever automated order dispatching until an engineer manually logged in to update routing tables. By utilizing dynamic routing protocols (like OSPF or BGP in their enterprise data centers), the network automatically reroutes live delivery telemetry and API traffic through alternate available cloud regions within milliseconds, ensuring uninterrupted order processing for customers.

Task 5: OSPF Sample Configuration & Line-by-Line Annotation

Open Shortest Path First (OSPF) is a link-state dynamic routing protocol widely used in enterprise networks. Below is a sample OSPF configuration for connecting two routers, followed by a detailed breakdown of each command.

Sample OSPF Configuration Script (Cisco IOS Style)

```
Router> enable
Router# configure terminal
Router(config)# router ospf 1
Router(config-router)# router-id 1.1.1.1
Router(config-router)# network 192.168.10.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.3.255 area 0
Router(config-router)# passive-interface GigabitEthernet0/0
Router(config-router)# end
Router# write memory
```

Line-by-Line Explanation & Annotation

- 1. router> enable / router# configure terminal** — Switches the router from User EXEC mode to Privileged EXEC mode, granting administrative access.
- 2. router(config)# router ospf 1** — Enables the OSPF routing process on the router with Process ID 1. The process ID is locally significant to the router and manages OSPF routing tables.
- 3. router(config-router)# router-id 1.1.1.1** — Assigns a unique 32-bit Router ID (1.1.1.1) to uniquely identify this router within the OSPF routing domain (often matching a loopback interface IP).
- 4. router(config-router)# network 192.168.10.0 0.0.0.255 area 0** — Advertises the local LAN subnet (192.168.10.0 with wildcard mask 0.0.0.255) into OSPF Area 0 (the Backbone Area). This tells OSPF to establish adjacencies and share routes for this network segment.
- 5. router(config-router)# network 10.0.0.0 0.0.3.255 area 0** — Advertises the point-to-point WAN link subnet (10.0.0.0 with wildcard mask 0.0.3.255) into Area 0, enabling the router to discover neighboring routers across the WAN.
- 6. router(config-router)# passive-interface GigabitEthernet0/0** — Prevents OSPF hello packets from being sent out of the specified interface (GigabitEthernet0/0 connected to end-user devices), saving bandwidth and securing local access while still advertising the network subnet.

7. router(config-router)# end & router# write memory — Exits configuration mode and saves the running configuration to NVRAM to ensure persistence across reboots.